

Predictive data analytics for contract renewals: a decision support tool for managerial decision-making

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Abstract

Purpose – Predictive analytics and artificial intelligence are perceived as significant drivers to improve organizational performance and managerial decision-making. Hiring employees and contract renewals are instances of managerial decision-making problems that can incur high financial costs and long-term impacts on organizational performance. The primary goal of this study is to identify the Major League Baseball (MLB) free agents who are likely to receive a contract.

Design/methodology/approach – This study used the design science research paradigm and the cognitive analytics management (CAM) theory to develop the research framework. A dataset on MLB's free agents between 2013 and 2017 was collected. A decision support tool was built using artificial neural networks.

Findings – There are clear links between a player's statistical performance and the decision of the player to sign a new offered contract. "Age," "Wins above Replacement" and "the team on which a player last played" are the most significant factors in determining if a player signs a new contract.

Originality/value – This paper applied analytical modeling to personnel decision-making using the design science paradigm and guided by CAM as the kernel theory. The study employed machine learning techniques, producing a model that predicts the probability of free agents signing a new contract. Also, a web-based tool was developed to help decision-makers in baseball front offices so they can determine which available free agents to offer contracts.

Keywords Machine learning, Sports analytics, Design science, Cognitive analytics management

Paper type Research paper

1. Introduction

Should a team renew a player's contract or let the player go? This is one of the most challenging questions MLB senior managers and executives face. In 2019, the total player payroll for the MLB was around \$3.9 billion (USA Today, 2019). The bulk of these contract renewals, however, are made on gut instinct (Laubert, 2019). Personnel employment decision-making is a surging challenge that has financial and strategic impacts across many industries (SIA, 2019). Over the last decades, academic studies have investigated the employment decision-making processes and called for more research to address the numerous research gaps and the growing needs of practitioners (Billsberry, 2007; Breaugh and Starke, 2000; Schnake, 2016; Vinayak et al., 2017). Predictive analytics and artificial intelligence (AI) are perceived as major drivers to improve organizational performance and managerial decision-making processes. Examining the significantly challenging case of MLB free agents' recruitment, this study aims to help managers in the personnel decision-making process. Specifically, the paper is constructing an innovative predictive framework (IT artifact) that uses predictive analytics and machine learning techniques to identify which players will sign a new contract. Also, this paper is employing the design science research paradigm to develop another innovative IT artifact – a web-based tool to calculate the probability of a player earning a new contract.



Based on our literature review as provided in [Section 3](#), and to the best of our knowledge, none of the prior studies attempt to model which free agents meet the criteria of signing a new contract. Therefore, the research objective of this study is to develop a model that predicts the probabilities of free agents signing a new contract. The framework can be applied in baseball front offices to help decision-makers identify players that should be targeted for acquisition.

The remainder of the paper is organized as follows. The second section introduces the research framework and kernel theory. The third section reviews the literature on MLB contracts and analytics. The fourth section outlines the proposed framework. The fifth section discusses the evaluation of the framework and the results. The sixth section presents the decision support tool. Finally, the conclusion is offered.

2. Research framework

To achieve our research objective, we follow the design science research paradigm to guide the construction of the IT artifact, as it is an overarching framework for developing IT artifacts ([Abbasi et al., 2012](#); [Hevner et al., 2004](#)). The IT artifact is “a thing that has, or can be transformed into, material existence as an artificially made object (e.g. model, instantiation) or process (e.g. method, software)” ([Gregor and Hevner, 2013](#), p. 341). Based on the design science research paradigm, this paper presents an innovative predictive framework utilizing state-of-the-art machine learning techniques and develops a decision support tool to aid managers in baseball front offices determine players that should be targeted for acquisition.

2.1 Kernel theory

Following the design science paradigm, the first step is to select a kernel theory. In identifying the relevant and salient factors associated with employment processes, we examine business analytics and human resource management theories as well as prior research that examined the employment and recruitment processes. In the business analytics field, we identify the innovative interdisciplinary Cognitive Analytics Management (CAM) theory as a kernel theory to guide the development of the predictive framework and predictive tool. CAM is an objective-driven paradigm that uses data to develop informed decisions that transform an organization to achieve its objectives ([Osman et al., 2019](#); [Osman and Anouze, 2017](#)). CAM comprises three processes: cognitive, analytic and management process. The cognitive process entails examining the organization’s challenge and “frames it into analytics terms by building data models, identifying variables and relationships to align with the desired goals/objectives which are set in coordination with senior management” ([Osman et al., 2019](#), p. 518). The analytics process explores the adequate analytics models and techniques to solve the problem and achieve the desired goal, as well as evaluating the results ([Osman et al., 2019](#)). The management process focuses on communicating the results and allocating the resources needed to achieve the targeted transformations ([Osman et al., 2019](#)). We propose that CAM is the most appropriate kernel theory for our study due to the interdisciplinary nature of the techniques needed to guide the design of our predictive model and predictive tool, as well as the need to utilize the three processes involved in CAM.

Based on CAM, the study should begin with the cognitive process. Therefore, we explore the required rationale to choose the features for the predictive model. Consequently, we identify the salient guidelines from the relevant human resource management literature and the Human Capital Theory. The seminal literature on Human Capital Theory is a stream of thousands of published papers across several research disciplines ([Sweetland, 1996](#)). According to Human Capital Theory, experience and on-the-job training improve the intangible human capital of the worker that, in turn, will increase worker’s productivity ([Becker, 1975](#); [Mincer, 1974](#); [Schultz, 1981](#)). In terms of baseball players, the features that are

related to skills, training and experience should be used to explore a player's performance and hence, the player's employment status. Following CAM, the analytics process and the management process will also guide the steps in the design of our IT artifacts.

3. Literature review

The sports industry in the United States is worth hundreds of billions of dollars (Heitner, 2015; Koseler and Stephan, 2017). When the market size of spectator sports is included, the industry is one of the top ten business domains around the globe (Fry and Ohlmann, 2012). Given the monetary value of the industry, it is not surprising that the use of analytics in the sports management field is becoming increasingly essential. The primary goal of sports analytics is to enable management to make better operational, tactical and strategic decisions through quantitative analyses and assessments of data. The application areas of sports analytics include analyzing players' performances, identifying draft-eligible players, quantifying the contribution of player types to team performance and scheduling players. However, with the high competitiveness in this multi-billion industry, some professional organizations have a strong incentive to keep their techniques proprietary (Koseler and Stephan, 2017).

The use of analytics in baseball began with the initiation of Sabermetrics research in the middle of the twentieth century. Sabermetrics refers to the empirical analysis of baseball with an emphasis on baseball statistics. James (1982) used box scores to compute various player statistics and made Sabermetrics known to the public. Bennett and Flueck (1983) used an explicit run production model and proposed several statistics to evaluate offensive performance in baseball. In 1984, New York Mets manager D. Johnson had a computer-based application created to compile and store metrics on team statistics (Porter, 1984). Retrosheet was founded in 1989 by D. Smith to computerize the box score of every major league baseball (MLB) game. In the 1990s, Oakland Athletics have used analytics to reduce payroll costs and achieve success in the American League (Howard, 2016). Through analytical methods, they have identified that players with better capability of getting on base in ways other than producing hits are valuable assets to the team and can be acquired cheaper in free agency or trades as they tend to be undervalued by MLB.

A study conducted by Krautmann and Donley (2009) proposed a stochastic productivity model to evaluate if long-term contracts diminish. Kahn (1993) explored work effort and performance in the professional baseball industry and modeled the impact of managerial quality on a team's winning and player performance using a regression analysis and an MLB dataset acquired between 1969 and 1987. The study found that higher-quality managers lead to higher winning percentages and better player performance. Kahn (1993a) utilized concepts from game theory and modeled the effects of eligibility for free agency or arbitration on salary and contract duration. Das and Das (1994) build a model to analyze which characteristics of a ball in flight contribute most to a fielder's ability to catch it. Charnes *et al.* (1994) proposed a model to evaluate players' performances using Data Envelopment Analysis (DEA). Similarly, Anderson and Sharp (1997) developed a model to assess batters' skills using DEA and Composite Batter Index (CBI).

Yang and Swartz (2004), using several parameters such as past performance and batting ability, built Bayesian models to compute the probability of a team winning a particular game. Jensen (2009) estimated posterior distributions of home run rates using hidden Markov Chains and then employed a Bayesian model to assess how the hitting performances of players were evolving over their careers. Huddleston (2012) used a Bayesian machine learning technique and predicted future performance in fantasy baseball. Herrlin (2015) built multiple Bayesian-based models and optimized the fantasy baseball roster. These models were then utilized to build regression trees and predict outcomes for the player throughout the season.

Ganeshapillai and Gutttag (2014) and Hoang *et al.* (2015) utilized machine learning techniques to determine whether a pitcher's next pitch will be a fastball. The algorithms used in these studies included various Support Vector Machine (SVM) models with different kernels (e.g. linear and Gaussian), k-nearest neighbors (KNN) and Linear Discriminant Analysis (LDA). Valero (2016), using ten years of historical data, compared the performance of several machine learning algorithms in terms of predicting the winner of a baseball game for all the teams in the MLB. Bock (2015) and Sidle and Tran (2018) built multi-class classification models employing multinomial logistic regression, SVM and LDA to determine the pitch type of non-fastball pitches (e.g. curveballs, sliders). These two studies also implemented feature selection techniques and identified the most important predictors as spin direction, spin rate, start speed and break angle. Analytics has also been effectively used for team-building strategy. For example, Barnes and Bjarnadóttir (2016) employed decision-tree learning to assess if MLB free agents are appropriately valued based on their performance in the upcoming season. The results indicate a significant correlation between the historical performance of players and the terms of their contracts while addressing that other factors such as age and name appeal play a secondary role in determining contract value. In a recent systematic literature review of machine learning applications in baseball, Koseler and Stephan (2017) found that prior research has dominantly utilized SVM and KNN algorithms on baseball datasets and is expected to rely more on neural networks.

According to our literature review, and to the best of our knowledge, none of the prior studies attempt to model which free agents will sign a new contract. Therefore, this study contributes to the extant body of the literature by developing machine learning models that predict the probabilities of free agents signing a new contract. In addition to the deployment of the machine learning algorithm, the importance of each variable from each model is extracted via Sensitivity Analysis (SA). The obtained importance values are then fused/combined via an Information Fusion (IF) technique. Finally, the best performing model is incorporated in a web-based prediction tool that helps baseball front offices to identify players that should be targeted for acquisition.

4. Proposed framework

Following the CAM analytics process, to identify whether a free agent will sign a new contract, we propose an analytical methodology consisting of three consecutive steps using machine learning. Figure 1 presents the outline of the proposed methodology. The first step preprocesses the dataset by handling categorical variables, imputing missing values and balancing the ratio of the output variable with random under-sampling. The second step models the data using three ML algorithms, namely Support Vector Machine (SVM), Artificial Neural Networks (ANN) and Random Forests (RF). This step optimizes the parameters of these aforementioned machine learning models using three-fold cross-validation and computes the relative importance of variables through sensitivity analysis. The relative importance values are combined via the Information Fusion (IF) technique to determine the best predictors that will yield desired accuracy rates. In Step 3, a decision support tool utilizing the selected machine learning algorithm is constructed to help decision-makers determine the likelihood of a player signing a new contract. This high-level tool enables management to make a strategic decision without expertise in statistics, machine learning and artificial intelligence.

4.1 Predictive models

We employ three different types of predictive models: ANNs (Abraham, 2005), SVM (Gunn, 1997) and RF (Breiman, 2001). These algorithms are selected based on our preliminary analysis.

Bagging is used to reduce the variance of an estimated prediction function, which has high variance and low bias (e.g. decision trees). RF is a version of bagging and uses an ensemble learning algorithm (Breiman, 2001). The original idea behind RF is to build an extensive

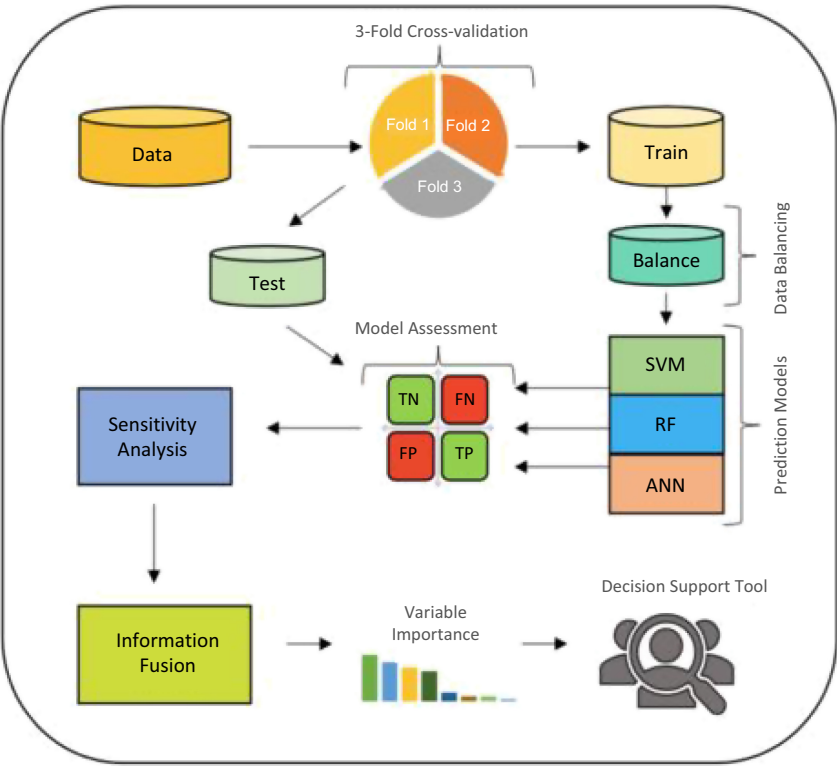


Figure 1.
Outline of the proposed
methodology

collection of “de-correlated” decision trees using random samples with replacement. RF aims to reduce the correlation between the trees by randomly selecting input variables during the tree-growing process. Decision trees in RF are constructed top-down by finding the attributes that best classifies the training examples. To identify which variables are the best classifiers in a given node, two statistical properties, called information gain or entropy, can be used. Entropy measures the impurity of the training examples. The information gain of a particular variable is the expected reduction in entropy caused by are partitioned the training examples based on this variable. By computing the information gain iteratively for training examples, the tree structure is augmented with branches. RF utilizes the same principles in decision trees and builds many trees with few branches, instead of a single tree with many branches. The ability of RF to prune trees before they become large prevents overfitting and reduces the bias. In the present study, the parameters of RF—the number of variables used for each individual tree and the number of trees grown—are tuned using k-fold cross-validation (see Figure 2).

SVM finds a hyperplane that divides the instances of separate classes by the widest gap. Support vectors are the data points located closest to this hyperplane. These data points are the most difficult instances to classify and directly affect the location of the decision surface.

SVM, using a kernel function, can also effectively perform nonlinear classification by implicitly conducting nonlinear transformation and mapping instances into high-dimensional feature spaces (Hastie *et al.*, 2009). In this study, a radial-based kernel function is used to develop the SVM model, and its performances (e.g. cost and sigma) are tuned to obtain an optimal model (see Figure 3).

ANN is a supervised machine learning algorithm that generates linear combinations of input variables as derived features. Then, the algorithm feeds these derived features through an activation function (e.g. sigmoid) to model the target variable. ANN has a structure with multiple layers that includes an input layer, a hidden layer and an output layer. This particular structure allows ANN to model highly nonlinear relationships (James *et al.*, 2013). Specifically, in this study, we adopt a vanilla structure to create the ANN models; namely, the single hidden layer back-propagation network was deployed. The vanilla model uses the back-propagation algorithm to find the solution to the optimal parameters. It should be noted that ANNs are prone to overfitting the data due to a high number of parameters that they involve. To mitigate this problem, the model is regularized by adding a penalty term on the parameter estimations that penalizes the loss function: known as weight decay. The other parameter of the ANN model—the number of units in the hidden layer—is tuned using cross-validation because there is no explicit way to select optimal parameters (see Figure 4).

Table 1 conceptually summarizes these aforementioned machine learning algorithms and provides their advantages and disadvantages.

4.2 Cross-validation

K -fold cross-validation is used to evaluate the performance of the models and identify the best model. This technique is particularly important as the size of the dataset used in this study is relatively small, and model overfitting is of an issue (Olson and Delen, 2008). In this procedure, the original dataset is randomly partitioned into k -subsets. A model is trained using the $k-1$ folds, and then it is tested on the fold that has not been used during the training

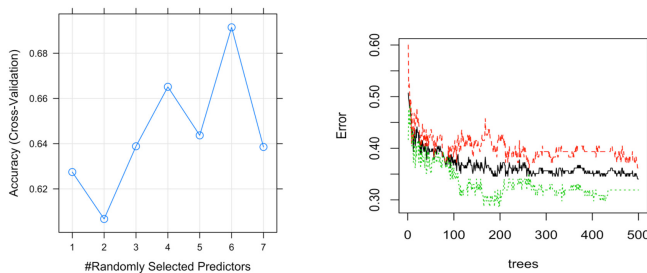


Figure 2.
Parameter tuning for
random forest

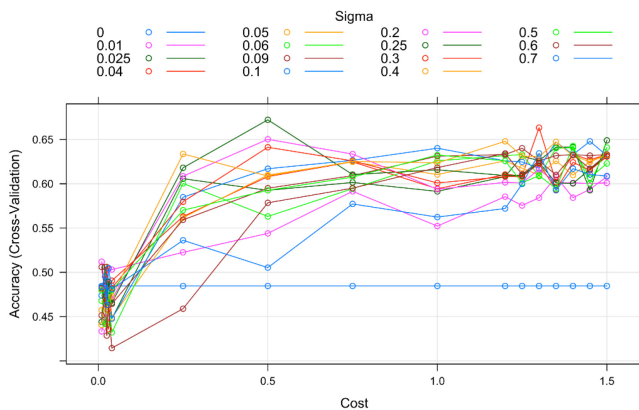
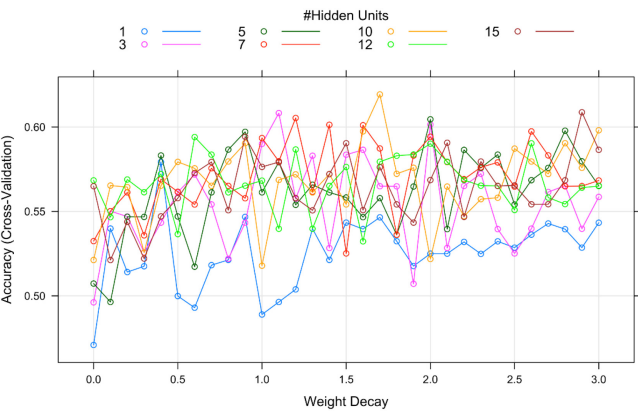


Figure 3.
Parameter tuning for
support vector
machine

Figure 4.
Parameter tuning
for ANN



Prediction model	Summary of advantages	Summary of disadvantages
RF	(1) Reduces variance (2) Identifies variable importance (3) Robust to outliers and nonlinear data (4) Simpler to tune as compared to SVM and ANN	(1) Has a “Blackbox” nature (2) Has low interpretability (3) Favors categorical variables with a higher number of levels
SVM	(1) Performs well with linear dependencies (2) Performs relatively better than RF with sparse data (3) Works well with different data types (e.g. semi-structured)	(1) May not work well with data containing many categorical variables (2) May be difficult to define the most appropriate distance function (3) Has many parameters to tune (e.g. type of kernel function and kernel specific parameters)
ANN	(1) Models complex nonlinear relationships (2) Does not impose assumptions regarding data (3) Can work with incomplete data	(1) Hard to interpret the model (2) May not perform well on small datasets (3) Prone to overfitting

Table 1.
Summary of ANN,
SVM and RF

phase. The formula for k -fold cross-validation is given in Equation (1), where P_i and k represent the performance measure of the i th fold and the total number of folds, respectively.

$$CVP = \frac{1}{k} \sum_{i=1}^k P_i \tag{1}$$

The pseudocode for k -fold cross-validation is as follows (Kohavi, 1995; Schneider, 1997)

- (1) Randomly Shuffle the dataset
- (2) Split the dataset into k folds
- (3) For each unique fold:
 - Use it as a hold-out set to test the model
 - Use the remaining folds as a training dataset

- Fit a model on the training set and evaluate it on the hold-out set
 - Retain performance indicators (e.g. AUC)
- (4) Summarize the models using the performance indicators

5. Evaluation

5.1 Data acquisition and exploration

As discussed in Section 2, CAM cognitive theory guides our selection of features in the model. The dataset was acquired from the <https://www.baseball-reference.com> Website, which monitors the transactions of the MLB and contains publicly available statistical data for current and historical baseball players, teams and scores. The dataset includes information regarding free agents that have left teams and new free agents that teams have signed to replace them between 2013 and 2017. Cleaning the data consisted of multiple steps: the first step of data cleaning was to eliminate observations that were outside the scope of this study. Pitchers, who made up half of the observations, were removed because of the contrast between hitters and pitchers statistically. There are few overlapping statistics between the two, and having both in the model would have skewed the perceived effectiveness of the variables. For example, pitchers, on average, have very low batting averages, and if a pitcher was resigned, that low batting average would impact the threshold for classification found by the model. The next step was the removal of observations with missing values. The dataset was mostly complete, so very few observations had to be removed due to missing values. The variables in this dataset and their brief explanations are provided in Table 2.

Missing data points were categorized to determine an appropriate technique to handle missing values. In machine learning, there are three categories of missing data, and they are defined as follows:

- (1) Missing completely at random (MCAR): The reason why a data point is missing is not due to its hypothetical value or its correlation to other variables.
- (2) Missing at random (MAR): The reason why a data point is missing is related to some of the observed data. In other words, the data is not missing due to its value.
- (3) Missing not at random (MNAR): A data point may be missing due to its value.

A hypothesis test proposed by Little (1988) to evaluate if the missing values were MCAR was conducted. The result indicated that the missing data were MCAR; therefore, the missing values, accounting for ~1.2% of the entire dataset, were removed using the list-wise deletion technique, yielding a dataset with 874 observations. Then the output variable indicating whether a player was reassigned was encoded using one-hot-encoding. The distribution of the output variable along with the summary statistics of the variables is provided in Table 3.

5.2 Results

The performance results of the aforementioned machine learning algorithms using the preprocessed MLB dataset are given in Figure 5 and Table 4. It is observed that the highest AUC score of 0.72 is obtained by the RF model, followed by the SVM model that achieves an AUC score of 0.699, and the ANN model that reaches an AUC score of 0.685. Similarly, RF model reaches the highest score of Accuracy (0.698) and Recall (0.715). With that said, the highest specificity score of 0.657 is obtained by ANN. However, as specificity, recall and accuracy metrics can be manipulated by altering the decision cutoff value, we opt to use AUC as the primary metric to decide on the best performing model, as it is not sensitive to a cutoff value. Therefore, RF model is chosen as the best performing model, and it is incorporated into the web-based prediction tool.

Table 2.
Definition of the
variables

Input variables	Explanation	Definition
Team	–	Name of the team for which the player played in the previous season
Age	–	Age of players on June 30th of that season – approximate midpoint chosen by Baseball-Reference
WAR	Total wins above replacement	Total WAR accumulated by a player over the previous three seasons. WAR is a metric used to evaluate a player against the average minor leaguer called up to spend time in MLB
G	Games	Total number of games that a player has played in his career
AB	At-Bats	Total number of plate appearances a player has had in his career that ended in either a hit or out. Does not include plate appearances ending in a walk, hit-by-pitch, sacrifice fly or bunt or interference
R	Runs	Total number of times a player has crossed home plate in their career. Usually higher for contact players than power hitters
H	Hits	Total number of times a player has put the ball in play (or home run) and reaches at least first base safely
HR	Homeruns	Total number of home runs hit by a player in their career
RBI	Runs batted in	Total number of times a player has caused a fellow player to reach home base as the result of a hit or sacrifice bunt or fly. Usually higher for power hitters than contact hitters
SB	Stolen bases	Total number of bases stolen by a player and occurs when a player advances bases when the ball is not put into play by a batter
BB	Bases on balls	Total number of times a player has reached base safely due to an accumulation of pitches thrown outside the strike zone
BA	Batting average	Career batting average of that player. Computed as Hits/At-bats
OBP	On base percentage	Career ratio of how often a player gets on base. Computed as (Hits + Walks + Reaches on Error)/Total Plate Appearances
SLG	Slugging percentage	Career metric that captures how much a player hit for power. Calculated as Total Bases (1*Number of Singles + 2*Number of Doubles + 3*Number of Triples + 4*Number of Home Runs)/At-bats
OPS	On base percentage	Career ratio of how often a player gets on base. Computed as (Hits + Walks + Reaches on Error)/Total Plate Appearances
Output Variable	Meaning	Explanation
Signed	–	Whether a player signed a new contract that following season. 1 indicates a new contract was signed by the player

The relative importance of each variable was extracted from the predictive models through Sensitivity Analysis (SA) (Saltelli, 2002). The SA technique measures the relative contribution of each variable to the model outcome. To do so, SA considers the ratio of the error of the model that includes a specific variable to the error of the model that excludes that specific variable. This ratio indicates to what extent much the model performance changes when a variable is excluded from the model, thereby providing the relative importance. We normalized the sensitivity of each variable for each predictive model as described by Saltelli *et al.* (2002) and displayed in Figure 6.

In order to increase the robustness and accuracy of the information provided by SA, IF technique is utilized. IF, by combining information from many different sources, decreases uncertainty and bias that may occur otherwise (Torra, 2003). The fused importance of variables, which obtained by combining individual importance values of the proposed SVM, ANN, RF algorithms via IF, is given in Figure 7.

The three variables with the highest level of importance values are “Age,” “Wins Above Replacement over the past three seasons (*war*),” and “*team* that the player last played on.” The variables identified as important in this study are aligned with the discussions in the

Categorical variable name	Number of categories		Distribution of categories	
Team	31		–	
Signed	2	No	Yes	
		138 (15.8%)	736 (84.2%)	
Continuous variable name	Mean	Median	Range	
AGE	31.43	31	24–44	
WAR	1.3	0.2	–4–22	
G	582.3	431	1–2,747	
AB	1879.4	1,152	1–11,195	
R	249.8	129.5	0–1923	
H	490.8	282.5	0–3,465	
HR	56.2	20.5	0–440	
RBI	234	113.5	0–1,587	
SB	36.08	10	0–614	
BB	173	93.5	0–1,476	
BA	0.2314	0.244	0–0.5710	
OBP	0.2922	0.308	0–0.6250	
SLG	0.3538	0.371	0–1.2860	
OPS	0.646	0.673	0–1.911	

Table 3.
Summary of the variables

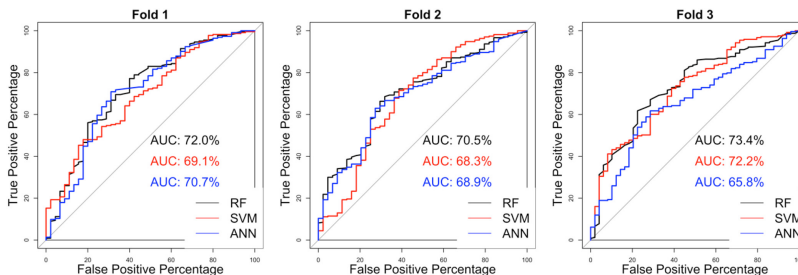


Figure 5.
ROC curves for predictive models

Model	Folds	Accuracy	AUC	Recall	Specificity
SVM	Fold 1	0.612	0.691	0.610	0.622
	Fold 2	0.637	0.683	0.637	0.636
	Fold 3	0.658	0.722	0.663	0.633
	Mean	0.636	0.699	0.637	0.630
	St. dev	0.023	0.021	0.027	0.007
RF	Fold 1	0.705	0.720	0.726	0.600
	Fold 2	0.678	0.705	0.678	0.682
	Fold 3	0.712	0.734	0.741	0.571
	Mean	<i>0.698</i>	<i>0.720</i>	<i>0.715</i>	<i>0.618</i>
	St. dev	0.018	0.015	0.033	0.058
ANN	Fold 1	0.631	0.707	0.610	0.733
	Fold 2	0.662	0.689	0.667	0.636
	Fold 3	0.620	0.658	0.714	0.601
	Mean	0.638	0.685	0.664	0.657
	St. dev	0.022	0.025	0.052	0.068

Note(s): The best classification results are in “Italic”

Table 4.
Classification results

literature. For example, according to Strober (1990), older players tend to stay in their teams due to decreasing athletic skills and not wanting to start their careers over.

After exploiting the relative variable importance. Partial dependence is employed to further explore the marginal effect of the most important three predictor variables on the target variable, which promotes the interpretability of complex machine learning algorithms and consequently increases the understanding of the relationship between the players' statistics and the chance of the players signing a new contract.

It can be seen from Figure 8 that the chance of free agents signing a contract with a team increases by the age of 32, being over 30% more likely to sign a new contract. The trend experience drastic drop down after approximately the age of 32 and hits being 10% less like to sign a new contract, and the trend remains the same with a slight decrease afterward. The partial dependence plot on Wins above replacement (WAR), the right top, indicates that growth in WAR increases the chance free agents sign a new contract. Lastly, it seems that the teams play an important role in free agents signing a new contract. For example, a free agent playing for Los Angeles Angles (ANA) is 60% more likely to sign a new contract. However, on the other hand, a free agent that plays for Miami Marlins (MIA) is deemed to 40% less likely to sign a new contract. However, it is noteworthy that the accuracy/reliability of these interpretations (dependence plots) heavily depend on the orthogonality assumption of the predictors, which is more than often violated in practice.

6. Decision support tool

Based on the CAM management process, the study proposes a tool to facilitate the implementation of the predictive model results. Therefore, we developed a graphical user interface, which is a web-based decision support tool wherein the best performing model – Random Forest – is incorporated. The tool is developed using the R programming language and can be accessed at https://research_tool/baseball/.

Figure 6.
Sensitivity analysis for
predictive models

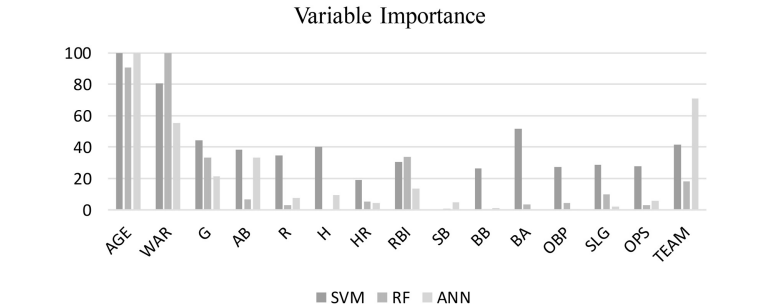


Figure 7.
Information fusion for
variable importance

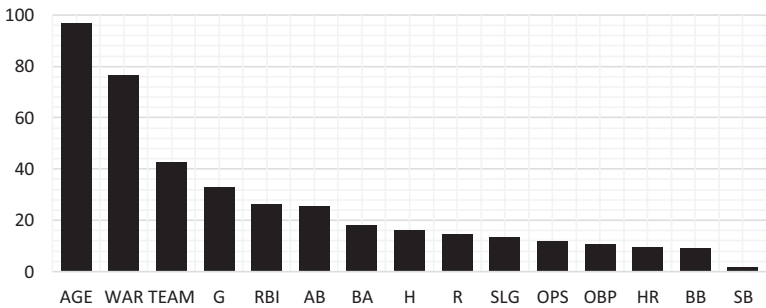




Figure 8.
Partial dependence plot

This tool receives the career statistics for a player as inputs. Then it calculates the probability of that player renewing a new contract, which enables teams' managers to make data-driven strategic decisions without having any expertise in analytics. Therefore, this artifact provides team managers with the necessary information in the timeliest and effective fashion

Age: 30	Free agent will sign? : YES
WAR: 12.5	Probability of renewing the contract is: 0.61 (61 %)
G: 1519	
AB: 6139	
R: 878	
H: 1311	
HR: 308	
RBI: 862	
SB: 60	
BB: 881	
BA: 0.255	
OBP: 0.368	
SLG: 0.493	
OPS: 0.861	
Team: PIT	
Submit	

Plate 1.
A screenshot of the
prediction tool

and helps them better identify players who are going to be free agents. Consequently, teams can save a significant amount of money as they can keep the free agents instead of replacing them with other players for higher costs. [Plate 1](#) shows the interface of the decision support tool.

7. Conclusion

This paper applied analytical modeling to personnel decision-making using the design science paradigm and guided by CAM as kernel theory. “Age” plays the most significant role in determining if a player can achieve a new contract. Additionally, “Wins Above Replacement” is a critical factor in identifying which players may receive a new contract. The models showed that the team on which a player last played impacts the team’s decision to sign that particular player. The findings in this analysis can be beneficial to members of MLB front offices as a tool to assist team building and roster management. Decision-makers can target which free agents meet the criteria of a player who earns a new contract, then adjust their talent acquisition strategy accordingly. Knowing who can still perform at an MLB caliber during the free-agency period helps reduce the risk for MLB teams, thus leading to a more efficient team structure for future seasons.

References

- Abbasi, A., Albrecht, C., Vance, A. and Hansen, J. (2012), “MetaFraud: a meta-learning framework for detecting financial fraud”, *MIS Quarterly*, Vol. 36 No. 4, p. 1293.
- Abraham, A. (2005), “Artificial neural networks”, *Handbook of Measuring System Design*, John Wiley & Sons, Chichester, doi: [10.1002/0471497398.mm421](https://doi.org/10.1002/0471497398.mm421).
- Anderson, T.R. and Sharp, G.P. (1997), “A new measure of baseball batters using DEA”, *Annals of Operations Research*, Vol. 73, pp. 141-155.
- Barnes, S.L. and Bjarnadóttir, M.V. (2016), “Great expectations: an analysis of major league baseball free agent performance”, *Statistical Analysis and Data Mining*, Vol. 9 No. 5, pp. 295-309, John Wiley and Sons Inc.
- Becker, J.D. (1975), “The phrasal lexicon”, in Schank, R.C. and Nash-Webber, B.L. (Eds), *Theoretical Issues in Natural Language Processing: An Interdisciplinary Workshop in Computational Linguistics, Psychology, Linguistics, Artificial Intelligence*, 10–13 June 1975, Cambridge, Massachusetts, Association for Computational Linguistics, Stroudsburg, PA, pp. 70-73.
- Bennett, J.M. and Flueck, J.A. (1983), “An evaluation of major league baseball offensive performance models”, *The American Statistician*, Vol. 37 No. 1, pp. 76-82, Taylor & Francis.
- Billsberry, J. (2007), *Experiencing Recruitment and Selection*, Wiley, Chichester.
- Bock, J. (2015), “Pitch sequence complexity and long-term pitcher performance”, *Sports*, Vol. 3 No. 1, pp. 40-55, MDPI AG.
- Breaugh, J.A. and Starke, M. (2000), “Research on employee recruitment: so many studies, so many remaining questions”, *Journal of Management*, Vol. 26 No. 3, pp. 405-434, Elsevier BV.
- Breiman, L. (2001), “Random forests”, *Machine Learning*, Vol. 45 No. 1, pp. 5-32, Kluwer Academic Publishers.
- Charnes, A., Cooper, W.W., Lewin, A.Y. and Seiford, L.M. (Eds) (1994), *Data Envelopment Analysis: Theory, Methodology, and Applications*, Kluwer, Boston.
- Das, S. and Das, R. (1994), “Using reinforcement learning to catch a baseball”, *Proceedings of 1994 IEEE International Conference on Neural Networks (ICNN'94)*, Vol. 5, IEEE, pp. 2808-2812.
- Fry, M.J. and Ohlmann, J.W. (2012), “Introduction to the special issue on analytics in sports, Part I: general sports applications”, *Interfaces*, Vol. 42 No. 2, pp. 105-108, INFORMS.

- Ganeshapillai, G. and Gutttag, J. (2014), "A data-driven method for in-game decision making in MLB", *MIT Sloan Sports Analytics Conference*, Moston, MA.
- Gregor, S. and Hevner, A.R. (2013), "Positioning and presenting design science research for maximum impact", *Management Information Systems Quarterly*, Vol. 37 No. 2, pp. 337-355.
- Gunn, S. (1997), "Support vector machines for classification and regression", *ISIS Technical Report*, Vol. 14, pp. 85-86.
- Hastie, T., Tibshirani, R. and Friedman, J. (2009), *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, Springer, New York City.
- Heitner, D. (2015), "Sports industry to reach \$73.5 billion by 2019", available at: <https://www.forbes.com/sites/darrenheitner/2015/10/19/sports-industry-to-reach-73-5-billion-by-2019/#55e032841b4b>.
- Herrlin, D.L. (2015), "Forecasting MLB performance utilizing a Bayesian approach in order to optimize a fantasy baseball draft", Ph.D thesis, San Diego State University, California.
- Hevner, A.R., March, S.T., Park, J. and Ram, S. (2004), "Design science in information systems Research", *MIS Quarterly*, Vol. 28 No. 1, pp. 75-105.
- Hoang, P., Hamilton, M., Murray, J., Stafford, C. and Tran, H. (2015), "A dynamic feature selection based LDA approach to baseball pitch prediction", *Lecture Notes in Computer Science*, Vol. 9441, pp. 125-137, Springer Verlag.
- Howard, H. (2016), "The Sabermetric revolution: assessing the growth of analytics in baseball Benjamin Baumer and Andrew Zimbalist. Philadelphia: university of Pennsylvania press, 2014", *Journal of American Culture*, Vol. 39 No. 1, pp. 129-130, Wiley-Blackwell.
- Huddleston, S.D. (2012), "Hitters vs. Pitchers: a comparison of fantasy baseball player performances using hierarchical bayesian models", Ph.D thesis, Brigham Young University-Provo, Utah.
- James, B. (1982), *The Bill James Baseball Abstract, 1982*, Ballantine Books, New York.
- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2013), *An Introduction to Statistical Learning*, Springer New York, New York, NY, doi: [10.1007/978-1-4614-7138-7](https://doi.org/10.1007/978-1-4614-7138-7).
- Jensen, F.V. (2009), "Bayesian networks", *Wiley Interdisciplinary Reviews: Computational Statistics*, Vol. 1 No. 3, pp. 307-315.
- Kahn, L.M. (1993a), "Managerial quality, team success, and individual player performance in major league baseball", *Industrial and Labor Relations Review*, Vol. 46 No. 3, p. 531, JSTOR.
- Kahn, L.M. (1993b), "Free agency, long-term contracts and compensation in major league baseball: estimates from panel data", *The Review of Economics and Statistics*, Vol. 75 No. 1, p. 157, JSTOR.
- Kohavi, R. (1995), "A study of cross-validation and bootstrap for accuracy estimation and model selection", *Proceedings of the Fourteenth International Joint Conference on Artificial Intelligence*, Morgan Kaufmann, San Francisco, CA, pp. 1137-1143.
- Koseler, K. and Stephan, M. (2017), "Machine learning applications in baseball: a systematic literature review", *Applied Artificial Intelligence*, Vol. 31 Nos 9-10, pp. 745-763, Taylor and Francis Inc.
- Krautmann, A.C. and Donley, T.D. (2009), "Shirking in major league baseball: revisited", *Journal of Sports Economics*, Vol. 10 No. 3, pp. 292-304.
- Lauber, S. (2019), "Bryce Harper or Manny Machado? Phillies' Rhys Hoskins has a 'gut feeling that we'll get one of them'", available at: <https://www.inquirer.com/phillies/phillies-bryce-harper-manny-machado-rhys-hoskins-free-agency-mlb-aaron-nola-20190122.html>.
- Little, R.J.A. (1988), "A test of missing completely at random for multivariate data with missing values", *Journal of the American Statistical Association*, Vol. 83 No. 404, pp. 1198-1202, Taylor & Francis.
- Mincer, J. (1974), *Schooling, Experience, and Earnings*, Columbia University Press, New York.

- Olson, D.L. and Delen, D. (2008), "Data mining process", *Advanced Data Mining Techniques*, Springer, Berlin, Heidelberg, doi: [10.1007/978-3-540-76917-0_2](https://doi.org/10.1007/978-3-540-76917-0_2).
- Osman, I.H. and Anouze, A.L. (2017), "A cognitive analytics management framework (CAM-part 1): SAMAS components, leadership, Frontier performance growth, and sustainable shared value", *Decision Management: Concepts, Methodologies, Tools, and Applications*, Vols 1-4, pp. 151-227, IGI Global.
- Osman, I.H., Anouze, A.L., Irani, Z., Lee, H., Medeni, T.D. and Weerakkody, V. (2019), "A cognitive analytics management framework for the transformation of electronic government services from users' perspective to create sustainable shared values", *European Journal of Operational Research*, Vol. 278 No. 2, pp. 514-532, Elsevier B.V.
- Porter, M. (1984), "The PC goes to bat", *PC Magazine*, p. 209.
- Saltelli, A., Tarantola, S., Campolongo, F. and Ratto, M. (2002), *Sensitivity Analysis in Practice*, John Wiley & Sons, Chichester, doi: [10.1002/0470870958](https://doi.org/10.1002/0470870958).
- Saltelli, A. (2002), "Making best use of model evaluations to compute sensitivity indices", *Computer Physics Communications*, Vol. 145 No. 2, pp. 280-297, North-Holland.
- Schnake, M. (2016), "An exploratory investigation of explanations for the relative effectiveness of employee recruitment methods", *American Journal of Management*, Vol. 16 No. 2, pp. 40-45, doi: [10.33423/AJMV16I2.1879](https://doi.org/10.33423/AJMV16I2.1879).
- Schneider, J. (1997), *Cross-validation. A Locally Weighted Learning Tutorial Using Vizier 1*, available at: <http://www.cs.cmu.edu/~schneide/psfiles/Tutorial.ps>.
- Schultz, T.W. (1981), *Investing in People: The Economics of Population Quality*, University of California Press, Berkeley, Los Angeles.
- SIA (2019), *Skills Gap Widening, More HR Execs Prefer Hiring Gig Workers: Survey*, Staffing Industry Analysts, Mountain View, CA.
- Sidle, G. and Tran, H. (2018), "Using multi-class classification methods to predict baseball pitch types", *Journal of Sports Analytics*, Vol. 4 No. 1, pp. 85-93, IOS Press.
- Strober, M.H. (1990), "Human capital theory: implications for HR managers", *Industrial Relations: A Journal of Economy and Society*, Vol. 29 No. 2, pp. 214-239.
- Sweetland, S.R. (1996), "Human capital theory: foundations of a field of inquiry", *Review of Educational Research*, Vol. 66 No. 3, pp. 341-359, American Educational Research Association.
- Torra, V. (2003), *Trends in Information Fusion in Data Mining*, Springer, Berlin, Heidelberg, pp. 1-6.
- USA today (2019), "MLB salaries", available at: <https://www.usatoday.com/sports/mlb/salaries/>.
- Valero, S.C. (2016), "Predicting win-loss outcomes in MLB regular season games-a comparative study using data mining methods", *International Journal of Computer Science in Sport*, Vol. 15 No. 2, pp. 91-112, International Association of Computer Science in Sport.
- Vinayak, P.C., Khan, B.M. and Jain, M.C. (2017), "Role of signalling theory in potential applicant attraction: an employer branding perspective", *International Journal of Emerging Research in Management and Technology*, Vol. 6 No. 4, pp. 230-233, Advance Academic Publisher.
- Yang, T.Y. and Swartz, T. (2004), "A two-stage Bayesian model for predicting winners in major league baseball", *Journal of Data Science*, Vol. 2, pp. 61-73.

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